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Testování efektu AI Foundational models na analýzu NGS dat DNA sekvenování

Diplomová práce

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Vedoucí práce: Mgr. Vojtěch Bystrý, Ph.D. Brno 2026

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Abstrakt

V této bakalářské/diplomové/rigorózní práci se věnujeme ...

Abstract

In this thesis we study ...

Místo tohoto listu vložte kopii oficiálního zadání práce bez podpisů.

Poděkování

Na tomto místě bych chtěl(-a) poděkovat ...

Prohlášení

Prohlašuji, že jsem svou diplomovou práci vypracoval samostatně pod vedením vedoucího práce s využitím informačních zdrojů, které jsou v práci citovány. Prohlašuji, že jsem pro stylistické úpravy textu využil nástroje AI a to plně v souladu s principy akademické integrity.

Brno 13. Května 2026

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Tomáš Vamberský

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List of abbreviations

- $\mathbb{C}$ množina všech komplexních čísel
- $\mathbb{R}$ množina všech reálných čísel
- $\mathbb{Z}$ množina všech celých čísel
- $\mathbb{N}$ množina všech přirozených čísel

Introduction

Chapter 1

Theoretical background

1.1 Transcription factors

1.1.1 Transcription factors and DNA binding

Transcription factors (TFs) are proteins that modulate transcription by binding DNA in a sequence-specific manner. The target sequences are typically motifs of 6–20 bp and are recognized with tolerance for sequence variation. They act through regulatory elements such as promoters, enhancers, and silencers, by recruiting the transcription machinery or modulating its activity. Most TFs share a modular architecture: a DNA-binding domain (DBD) that contacts the target sequence, paired with one or more effector domains responsible for transactivation, repression, or dimerisation. Because the DBD determines binding specificity, TFs are usually classified by DBD family rather than by their effector domains or biological function. The human genome encodes approximately 1600 sequence-specific TFs, which can be classified into roughly 75 DBD families (Lambert et al., 2018). These families include the C2H2 zinc fingers (the family most represented in humans, e.g. CTCF), the basic helix-loop-helix proteins (e.g. MYC, which binds as a heterodimer with MAX), the ETS domain family (e.g. SPI1), and the forkhead family (e.g. FOXK2).

Despite the considerable variety in structure, all sequence-specific TFs face the same problem - they must recognise their short target sequence among numerous similar but non-functional sites in a genome.

1.1.2 Sequence determinants of binding

The specificity of protein-DNA binding is based on two complementary mechanisms: direct chemical recognition of bases (base readout), and recognition of sequence-dependent DNA geometry (shape readout).

Base readout works through direct contact, either through hydrogen bonds or van der Waals interactions, of amino acid side chains and the Watson-Crick edges of bases, predominantly in the major groove. The major groove presents a distinctive pattern of hydrogen bond donors and acceptors that allows discrimination between base pairs, including for example the orientation of A-T or T-A (Luscombe et al., 2001; Seeman

et al., 1976). The canonical example of base readout is the C2H2 zinc finger, in which each $\beta\beta\alpha$ module contacts approximately three base pairs via two or three specificity residues on the recognition helix, contributing additively to the binding affinity of the multi-finger array. Position weight matrices (PWMs) are the natural statistical encoding of this mechanism, treating each position as an independent contribution to binding affinity.

Unlike base readout, shape readout depends on sequence context beyond the contact position - the geometry at any given base pair is influenced by neighbouring bases over several positions in either direction. The relevant geometric features (minor groove width, propeller twist, helical twist, roll, and slide...) describe the orientation and stacking of base pairs along the helix and are not direct chemical signals of base identity. They are read by protein contacts that respond to local shape and electrostatics rather than to specific bases. The canonical example is recognition of narrow minor grooves by arginine residues: A-tracts (runs of A-T base pairs) produce locally narrowed minor grooves with concentrated negative electrostatic potential, which arginines insert into and stabilise without forming base-specific hydrogen bonds (Rohs et al., 2009).

Even taken together, base and shape readout cannot fully account for the specificity of in vivo TF binding. Both mechanisms are short-range: base readout depends on direct contacts within the binding footprint of typically 6–20 bp, and shape readout, while sensitive to surrounding sequence, is mediated through contacts within and immediately flanking that footprint (Yang et al., 2017). Yet these mechanisms are genuinely complementary rather than redundant. Crystal structures of TF–DNA complexes routinely show major-groove base-specific contacts and minor-groove shape-based contacts within the same interface, and quantitative analyses show that augmenting PWM-based models with shape features systematically improves binding prediction across TF families (Mathelier et al., 2016). *The limitation is not that the two mechanisms fail individually, but that their combined reach is bounded by the binding footprint. The discrepancy between motif occurrences in the genome and the subset that is actually bound — addressed in the next paragraph — therefore demands explanations beyond the recognition mechanisms reviewed here.*

Sequence-specific TFs recognise short DNA motifs, typically 6–20 bp long. The recognition is degenerate: most positions tolerate multiple bases, with information formally captured by position weight matrices [TODO: odkaz na dalsi kapitolu? cut protoze se opakuji?]. The motif as a whole specifies the binding sequence much less precisely than its nominal length would suggest. Sequence recognition mechanisms reviewed above operate over tens of base pairs at most. In the context of the 3 Gb human genome, hundreds of thousands to millions of matches are expected by chance alone. ChIP-seq, the standard assay for in vivo TF occupancy (described in the next section), typically identifies between a few thousand and a few tens of thousands of bound sites per TF per cell line. CTCF in K562 cells, for example, has approximately 52,000 IDR-thresholded ENCODE peaks, against approximately 870,000 matches to the canonical CTCF motif in the hg19 genome at standard FIMO thresholds (stricter thresholds reduce this to a few tens of thousands, but at the cost of missing many genuine binding sites) (Dozmorov et al., 2022). This gap between motif presence and actual binding has been termed the transcription factor specificity paradox, *originally formalised for prokaryotic TFs as*

the futility theorem by Wunderlich a Mirny (2009): in a sufficiently large genome, even highly informative motifs predict more matches than there are TF molecules in the cell, making purely sequence-based recognition statistically insufficient. In practice, this is reflected in the weak correlation between intrinsic in vitro affinity (measured by HT-SELEX or protein-binding microarrays) and in vivo occupancy (measured by ChIP-seq): high-affinity sites can remain unbound, while low-affinity sites are often bound (Slatery et al., 2014). This gap is partially explained by several mechanisms, each a substantial research field in its own right.

Chromatin accessibility is regarded as the dominant factor: only a small fraction of the genome may be accessed by proteins at any given time, the rest made unavailable by nucleosomes or compacted into heterochromatin. ATAC-seq, DNase-seq, and FAIRE-seq are used to identify accessible , and most in vivo TF binding occurs within it (Thurman et al., 2012). *Cooperative binding with cofactors is the second. Many TFs bind productively only as part of a complex — homodimers, heterodimers, or larger assemblies — so combinatorics restrict where binding can occur (Spitz a Furlong, 2012)[nemam access, najit lepsi].* MYC, for example, binds as an obligate heterodimer with MAX. Cellular context further narrows the set: TF abundance, post-translational modification, and subcellular localisation are all regulated and condition-dependent.

The fourth and final mechanism - sequence context beyond the motif - is the focus of this thesis. Flanking nucleotides, dinucleotide composition, DNA shape, and broader compositional features of the regions surrounding a motif modulate binding affinity and contribute to the discrimination between bound and unbound matches (Dror et al., 2015). Unlike chromatin and cooperativity, this contribution is encoded directly in the DNA sequence and is therefore in principle accessible to any model that operates on raw sequence input. The next paragraph develops what is currently known about this contribution, from the immediate flanks of the motif out to the kilobase-scale compositional gradients recently characterised by Faltejsková et al. (2026) — the biological signal that motivates the rest of this thesis.

The fact that motif occurrences vastly outnumber bound sites motivates a substantial body of work on what else in the DNA sequence influences in vivo binding. The contributions that have been characterised span three orders of magnitude in distance.

At the shortest scale, the nucleotides immediately flanking the core motif modify binding affinity in ways that PWMs cannot capture. (Dror et al., 2015) showed that flanking-region features improve TF binding prediction across diverse families, with much of the gain attributable to DNA shape: the flanks determine the local geometry of the motif region itself, and so contribute through shape readout rather than through new direct contacts. At the intermediate scale of tens to hundreds of base pairs, *broader compositional features modulate the nonspecific affinity that governs how a TF samples DNA during 1D diffusion along the helix.* [TODO:check for phrasing in the paper] Sequence correlations and poly(dA:dT) tract content shape this nonspecific affinity landscape (Afek a Lukatsky, 2013; Sela a Lukatsky, 2011), so the sequence environment a scanning TF traverses is not energetically flat. Berg et al. (1981) established that TFs locate their targets faster than three-dimensional diffusion would allow by alternating 3D excursions with 1D sliding along DNA, and Slutsky a Mirny (2004) formalised the *kinetic trade-off between fast scanning and stable specific binding*. A target search of this

kind is in principle sensitive to sequence-encoded biases over distances much longer than the binding site itself.

Faltejsková et al. (2026) recently characterised such biases at kilobase scale. In a systematic analysis of in vivo binding sites for ten TFs from ENCODE ChIP-seq, they identified a statistically significant enrichment of GC content spanning approximately 1–2 kb around the binding site for nearly all TFs studied, with magnitudes on the order of ten percentage points [numbers?] and largely symmetric profiles around the peak centre. For a subset of TFs — notably MYC — they additionally observed directional dinucleotide asymmetries: upstream-enriched AA, CA and AC dinucleotides mirrored by downstream-enriched TT, TG and GT, producing a sequence gradient pointing toward the binding site. From these observations the authors propose a statistical funnel model in which broad compositional gradients create a low-free-energy landscape biasing 1D-scanning TFs toward their targets; once near the binding site, short-range readout completes the binding event. A theoretical feature of this proposal is that the gradients extend well beyond the *Debye screening length* over which direct electrostatic protein–DNA interactions decay, so the signal cannot be read out by long-range electrostatic contact and must instead modulate the scanning process itself. This study is *purely descriptive* — the authors propose the funnel as a hypothesis consistent with their observations rather than as a validated mechanism — and they do not test whether any model trained on raw sequence implicitly encodes the gradients they describe. That question motivates the rest of this thesis.

1.1.3 Experimental mapping of binding sites

Throughout this thesis, "binding site" refers to a position identified by ChIP-seq, the standard high-throughput assay for in vivo TF occupancy. The procedure consists of cross-linking protein–DNA contacts in living cells (typically with formaldehyde), fragmenting the chromatin, immunoprecipitating fragments bound by the TF of interest using a specific antibody, and sequencing the recovered DNA. Reads are aligned to the reference genome, where they accumulate as enrichment peaks at bound regions relative to a control library of input chromatin or non-specific IgG immunoprecipitate. Peak calling — the identification of statistically enriched regions against the local background — is conventionally performed with MACS2 (Zhang et al., 2008), and the output is a narrowPeak BED file containing the genomic coordinates of each peak, a significance score, and the offset of the position of maximum read pile-up within the peak (the summit).

Single-replicate peak calls contain false positives. The Irreproducible Discovery Rate framework (Li et al., 2011) addresses this by comparing peak rankings between biological replicates and retaining only peaks that are reproducibly called above a defined irreproducibility threshold. ENCODE provides IDR-thresholded peak files as the standard high-confidence output of its uniform processing pipeline, and these are the files used throughout this thesis as well as by Faltejsková et al. (2026).

A ChIP-seq peak is a region of read enrichment, typically a few hundred base pairs wide, rather than a single binding event. The actual point of TF–DNA contact is somewhere within the peak — most defensibly the summit, more conventionally the peak

midpoint. The choice between the two is methodological rather than substantive; *this thesis follows Faltejsková et al. in using the midpoint as the binding-site proxy, and the consequences of the choice are revisited in [odkazMetody?]*. Beyond this positional uncertainty, ChIP-seq has further well-known limitations: antibody quality varies between targets, cross-linking can capture transient or indirect contacts, fragment-length resolution caps the spatial precision at a few hundred base pairs, and any peak set is specific to the cell line and condition in which it was generated.

ENCODE is the data source for the analyses in this thesis. The project provides standardised processing, IDR-thresholded peak files, audit metadata for problematic experiments, and broad coverage of TFs across human cell lines, which together make it the natural reference resource for systematic TF studies — and the reason both Faltejsková et al. and this thesis use it.

1.2 Computational methods for motif analysis

1.2.1 PWMs

1.3 Foundation models in genomics

1.4 NGS data

Chapter 2

Aims of thesis

- 1.

Chapter 3

Methods $\int f(x) dx$ v názvu

$$\int f(x) dx \quad (\text{rovnice})$$

odkaz na rovnici s „tagem“ ([rovnice](#)) – pokud nechcete vzorci přidělit číslo, ale nějaký vlastní symbol, použijte „hvězdičková“ prostředí, tj. např. `equation*`

$$\iint f(x) dx \quad (3.1)$$

odkaz na druhý vzorec ([4.1](#))

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3.1 Podkapitola

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Phasellus placerat vulputate quam. Maecenas at tellus. Pellentesque neque diam, dignissim ac, venenatis vitae, consequat ut, lacus. Nam nibh. Vestibulum fringilla arcu mollis arcu. Sed et turpis. Donec sem tellus, volutpat et, varius eu, commodo sed, lectus. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Quisque enim arcu, suscipit nec, tempus at, imperdiet vel, metus. Morbi volutpat purus at erat. Donec dignissim, sem id semper tempus, nibh massa eleifend turpis, sed pellentesque wisi purus sed libero. Nullam lobortis tortor vel risus. Pellentesque consequat nulla eu tellus. Donec velit. Aliquam fermentum, wisi ac rhoncus iaculis, tellus nunc malesuada orci, quis volutpat dui magna id mi. Nunc vel ante. Duis vitae lacus. Cras nec ipsum.

Druhá kapitola s matematikou $\int f(x) dx$ v názvu

$$\int f(x) \mathrm{d}x \quad (\text{rovnice})$$
$$\iint f(x) \mathrm{d}x \quad (4.1)$$

Morbi luctus, wisi viverra faucibus pretium, nibh est placerat odio, nec commodo wisi enim eget quam. Quisque libero justo, consecetur a, feugiat vitae, porttitor eu, libero. Suspendisse sed mauris vitae elit sollicitudin malesuada. Maecenas ultricies eros sit amet ante. Ut venenatis velit. Maecenas sed mi eget dui varius euismod. Phasellus aliquet volutpat odio. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque sit amet pede ac sem eleifend consecetur. Nullam elementum, urna vel imperdiet sodales, elit ipsum pharetra ligula, ac pretium ante justo a nulla. Curabitur tristique arcu eu metus. Vestibulum lectus. Proin mauris. Proin eu

nunc eu urna hendrerit faucibus. Aliquam auctor, pede consequat laoreet varius, eros tellus scelerisque quam, pellentesque hendrerit ipsum dolor sed augue. Nulla nec lacus.

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetur odio sem sed wisi.

Sed feugiat. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Ut pellentesque augue sed urna. Vestibulum diam eros, fringilla et, consectetur eu, nonummy id, sapien. Nullam at lectus. In sagittis ultrices mauris. Curabitur malesuada erat sit amet massa. Fusce blandit. Aliquam erat volutpat. Aliquam euismod. Aenean vel lectus. Nunc imperdiet justo nec dolor.

Etiam euismod. Fusce facilisis lacinia dui. Suspendisse potenti. In mi erat, cursus id, nonummy sed, ullamcorper eget, sapien. Praesent pretium, magna in eleifend egestas, pede pede pretium lorem, quis consectetur tortor sapien facilisis magna. Mauris quis magna varius nulla scelerisque imperdiet. Aliquam non quam. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl.

Etiam ac leo a risus tristique nonummy. Donec dignissim tincidunt nulla. Vestibulum rhoncus molestie odio. Sed lobortis, justo et pretium lobortis, mauris turpis condimentum augue, nec ultricies nibh arcu pretium enim. Nunc purus neque, placerat id, imperdiet sed, pellentesque nec, nisl. Vestibulum imperdiet neque non sem accumsan laoreet. In hac habitasse platea dictumst. Etiam condimentum facilisis libero. Suspendisse in elit quis nisl aliquam dapibus. Pellentesque auctor sapien. Sed egestas sapien nec lectus. Pellentesque vel dui vel neque bibendum viverra. Aliquam porttitor nisl nec pede. Proin mattis libero vel turpis. Donec rutrum mauris et libero. Proin euismod porta felis. Nam lobortis, metus quis elementum commodo, nunc lectus elementum mauris, eget vulputate ligula tellus eu neque. Vivamus eu dolor. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Aenean nonummy turpis id odio. Integer euismod imperdiet turpis. Ut nec leo nec diam imperdiet lacinia. Etiam eget lacus eget mi ultricies posuere. In placerat tristique tortor. Sed porta vestibulum metus. Nulla iaculis sollicitudin pede. Fusce luctus tellus in dolor. Curabitur auctor velit a sem. Morbi sapien. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Donec adipiscing urna vehicula nunc. Sed ornare leo in leo. In rhoncus leo ut dui. Aenean dolor quam, volutpat nec, fringilla id, consectetur vel, pede.

Nulla malesuada risus ut urna. Aenean pretium velit sit amet metus. Duis iaculis. In

hac habitasse platea dictumst. Nullam molestie turpis eget nisl. Duis a massa id pede dapibus ultricies. Sed eu leo. In at mauris sit amet tortor bibendum varius. Phasellus justo risus, posuere in, sagittis ac, varius vel, tortor. Quisque id enim. Phasellus consequat, libero pretium nonummy fringilla, tortor lacus vestibulum nunc, ut rhoncus ligula neque id justo. Nullam accumsan euismod nunc. Proin vitae ipsum ac metus dictum tempus. Nam ut wisi. Quisque tortor felis, interdum ac, sodales a, semper a, sem. Curabitur in velit sit amet dui tristique sodales. Vivamus mauris pede, lacinia eget, pellentesque quis, scelerisque eu, est. Aliquam risus. Quisque bibendum pede eu dolor.

Donec tempus neque vitae est. Aenean egestas odio sed risus ullamcorper ullamcorper. Sed in nulla a tortor tincidunt egestas. Nam sapien tortor, elementum sit amet, aliquam in, porttitor faucibus, enim. Nullam congue suscipit nibh. Quisque convallis. Praesent arcu nibh, vehicula eget, accumsan eu, tincidunt a, nibh. Suspendisse vulputate, tortor quis adipiscing viverra, lacus nibh dignissim tellus, eu suscipit risus ante fringilla diam. Quisque a libero vel pede imperdiet aliquet. Pellentesque nunc nibh, eleifend a, consequat consequat, hendrerit nec, diam. Sed urna. Maecenas laoreet eleifend neque. Vivamus purus odio, eleifend non, iaculis a, ultrices sit amet, urna. Mauris faucibus odio vitae risus. In nisl. Praesent purus. Integer iaculis, sem eu egestas lacinia, lacus pede scelerisque augue, in ullamcorper dolor eros ac lacus. Nunc in libero.

Fusce suscipit cursus sem. Vivamus risus mi, egestas ac, imperdiet varius, faucibus quis, leo. Aenean tincidunt. Donec suscipit. Cras id justo quis nibh scelerisque dignissim. Aliquam sagittis elementum dolor. Aenean consectetur justo in pede. Curabitur ullamcorper ligula nec orci. Aliquam purus turpis, aliquam id, ornare vitae, porttitor non, wisi. Maecenas luctus porta lorem. Donec vitae ligula eu ante pretium varius. Proin tortor metus, convallis et, hendrerit non, scelerisque in, urna. Cras quis libero eu ligula bibendum tempor. Vivamus tellus quam, malesuada eu, tempus sed, tempor sed, velit. Donec lacinia auctor libero.

Praesent sed neque id pede mollis rutrum. Vestibulum iaculis risus. Pellentesque lacus. Ut quis nunc sed odio malesuada egestas. Duis a magna sit amet ligula tristique pretium. Ut pharetra. Vestibulum imperdiet magna nec wisi. Mauris convallis. Sed accumsan sollicitudin massa. Sed id enim. Nunc pede enim, lacinia ut, pulvinar quis, suscipit semper, elit. Cras accumsan erat vitae enim. Cras sollicitudin. Vestibulum rutrum blandit massa.

Sed gravida lectus ut purus. Morbi laoreet magna. Pellentesque eu wisi. Proin turpis. Integer sollicitudin augue nec dui. Fusce lectus. Vivamus faucibus nulla nec lacus. Integer diam. Pellentesque sodales, enim feugiat cursus volutpat, sem mauris dignissim mauris, quis consequat sem est fermentum ligula. Nullam justo lectus, condimentum sit amet, posuere a, fringilla mollis, felis. Morbi nulla nibh, pellentesque at, nonummy eu, sollicitudin nec, ipsum. Cras neque. Nunc augue. Nullam vitae quam id quam pulvinar blandit. Nunc sit amet orci. Aliquam erat elit, pharetra nec, aliquet a, gravida in, mi. Quisque urna enim, viverra quis, suscipit quis, tincidunt ut, sapien. Cras placerat consequat sem. Curabitur ac diam. Curabitur diam tortor, mollis et, viverra ac, tempus vel, metus.

Curabitur ac lorem. Vivamus non justo in dui mattis posuere. Etiam accumsan ligula id pede. Maecenas tincidunt diam nec velit. Praesent convallis sapien ac est. Aliquam ullamcorper euismod nulla. Integer mollis enim vel tortor. Nulla sodales placerat nunc.

Sed tempus rutrum wisi. Duis accumsan gravida purus. Nunc nunc. Etiam facilisis dui eu sem. Vestibulum semper. Praesent eu eros. Vestibulum tellus nisl, dapibus id, vestibulum sit amet, placerat ac, mauris. Maecenas et elit ut erat placerat dictum. Nam feugiat, turpis et sodales volutpat, wisi quam rhoncus neque, vitae aliquam ipsum sapien vel enim. Maecenas suscipit cursus mi.

4.1 Podkapitola

Quisque facilisis auctor sapien. Pellentesque gravida hendrerit lectus. Mauris rutrum sodales sapien. Fusce hendrerit sem vel lorem. Integer pellentesque massa vel augue. Integer elit tortor, feugiat quis, sagittis et, ornare non, lacus. Vestibulum posuere pellentesque eros. Quisque venenatis ipsum dictum nulla. Aliquam quis quam non metus eleifend interdum. Nam eget sapien ac mauris malesuada adipiscing. Etiam eleifend neque sed quam. Nulla facilisi. Proin a ligula. Sed id dui eu nibh egestas tincidunt. Suspendisse arcu.

Maecenas dui. Aliquam volutpat auctor lorem. Cras placerat est vitae lectus. Curabitur massa lectus, rutrum euismod, dignissim ut, dapibus a, odio. Ut eros erat, vulputate ut, interdum non, porta eu, erat. Cras fermentum, felis in porta congue, velit leo facilisis odio, vitae consectetur lorem quam vitae orci. Sed ultrices, pede eu placerat auctor, ante ligula rutrum tellus, vel posuere nibh lacus nec nibh. Maecenas laoreet dolor at enim. Donec molestie dolor nec metus. Vestibulum libero. Sed quis erat. Sed tristique. Duis pede leo, fermentum quis, consectetur eget, vulputate sit amet, erat.

Donec vitae velit. Suspendisse porta fermentum mauris. Ut vel nunc non mauris pharetra varius. Duis consequat libero quis urna. Maecenas at ante. Vivamus varius, wisi sed egestas tristique, odio wisi luctus nulla, lobortis dictum dolor ligula in lacus. Vivamus aliquam, urna sed interdum porttitor, metus orci interdum odio, sit amet euismod lectus felis et leo. Praesent ac wisi. Nam suscipit vestibulum sem. Praesent eu ipsum vitae pede cursus venenatis. Duis sed odio. Vestibulum eleifend. Nulla ut massa. Proin rutrum mattis sapien. Curabitur dictum gravida ante.

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Závěr

[illegible][illegible]

Quisque facilisis auctor sapien. Pellentesque gravida hendrerit lectus. Mauris rutrum sodales sapien. Fusce hendrerit sem vel lorem. Integer pellentesque massa vel augue. Integer elit tortor, feugiat quis, sagittis et, ornare non, lacus. Vestibulum posuere pellentesque eros. Quisque venenatis ipsum dictum nulla. Aliquam quis quam non metus eleifend interdum. Nam eget sapien ac mauris malesuada adipiscing. Etiam eleifend neque sed quam. Nulla facilisi. Proin a ligula. Sed id dui eu nibh egetas tincidunt. Suspendisse arcu.

Maecenas dui. Aliquam volutpat auctor lorem. Cras placerat est vitae lectus. Curabitur massa lectus, rutrum euismod, dignissim ut, dapibus a, odio. Ut eros erat, vulputate ut, interdum non, porta eu, erat. Cras fermentum, felis in porta congue, velit leo facilisis odio, vitae consectetur lorem quam vitae orci. Sed ultrices, pede eu placerat auctor, ante ligula rutrum tellus, vel posuere nibh lacus nec nibh. Maecenas laoreet dolor at enim. Donec molestie dolor nec metus. Vestibulum libero. Sed quis erat. Sed tristique. Duis pede leo, fermentum quis, consectetur eget, vulputate sit amet, erat.

Donec vitae velit. Suspendisse porta fermentum mauris. Ut vel nunc non mauris pharetra varius. Duis consequat libero quis urna. Maecenas at ante. Vivamus varius, wisi sed egestas tristique, odio wisi luctus nulla, lobortis dictum dolor ligula in lacus. Vivamus aliquam, urna sed interdum porttitor, metus orci interdum odio, sit amet euismod lectus felis et leo. Praesent ac wisi. Nam suscipit vestibulum sem. Praesent eu ipsum vitae pede cursus venenatis. Duis sed odio. Vestibulum eleifend. Nulla ut massa. Proin rutrum mattis sapien. Curabitur dictum gravida ante.

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nec, tempus at, imperdiet vel, metus. Morbi volutpat purus at erat. Donec dignissim, sem id semper tempus, nibh massa eleifend turpis, sed pellentesque wisi purus sed libero. Nullam lobortis tortor vel risus. Pellentesque consequat nulla eu tellus. Donec velit. Aliquam fermentum, wisi ac rhoncus iaculis, tellus nunc malesuada orci, quis volutpat dui magna id mi. Nunc vel ante. Duis vitae lacus. Cras nec ipsum.

Morbi nunc. Aliquam consectetur varius nulla. Phasellus eros. Cras dapibus portitor risus. Maecenas ultrices mi sed diam. Praesent gravida velit at elit vehicula portitor. Phasellus nisl mi, sagittis ac, pulvinar id, gravida sit amet, erat. Vestibulum est. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur id sem elementum leo rutrum hendrerit. Ut at mi. Donec tincidunt faucibus massa. Sed turpis quam, sollicitudin a, hendrerit eget, pretium ut, nisl. Duis hendrerit ligula. Nunc pulvinar congue urna.

Nunc velit. Nullam elit sapien, eleifend eu, commodo nec, semper sit amet, elit. Nulla lectus risus, condimentum ut, laoreet eget, viverra nec, odio. Proin lobortis. Curabitur dictum arcu vel wisi. Cras id nulla venenatis tortor congue ultrices. Pellentesque eget pede. Sed eleifend sagittis elit. Nam sed tellus sit amet lectus ullamcorper tristique. Mauris enim sem, tristique eu, accumsan at, scelerisque vulputate, neque. Quisque lacus. Donec et ipsum sit amet elit nonummy aliquet. Sed viverra nisl at sem. Nam diam. Mauris ut dolor. Curabitur ornare tortor cursus velit.

Morbi tincidunt posuere arcu. Cras venenatis est vitae dolor. Vivamus scelerisque semper mi. Donec ipsum arcu, consequat scelerisque, viverra id, dictum at, metus. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut pede sem, tempus ut, portitor bibendum, molestie eu, elit. Suspendisse potenti. Sed id lectus sit amet purus faucibus vehicula. Praesent sed sem non dui pharetra interdum. Nam viverra ultrices magna.

Aenean laoreet aliquam orci. Nunc interdum elementum urna. Quisque erat. Nullam tempor neque. Maecenas velit nibh, scelerisque a, consequat ut, viverra in, enim. Duis magna. Donec odio neque, tristique et, tincidunt eu, rhoncus ac, nunc. Mauris malesuada malesuada elit. Etiam lacus mauris, pretium vel, blandit in, ultricies id, libero. Phasellus bibendum erat ut diam. In congue imperdiet lectus.

Aenean scelerisque. Fusce pretium portitor lorem. In hac habitasse platea dictumst. Nulla sit amet nisl at sapien egestas pretium. Nunc non tellus. Vivamus aliquet. Nam adipiscing euismod dolor. Aliquam erat volutpat. Nulla ut ipsum. Quisque tincidunt auctor augue. Nunc imperdiet ipsum eget elit. Aliquam quam leo, consectetur non, ornare sit amet, tristique quis, felis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque interdum quam sit amet mi. Pellentesque mauris dui, dictum a, adipiscing ac, fermentum sit amet, lorem.

Ut quis wisi. Praesent quis massa. Vivamus egestas risus eget lacus. Nunc tincidunt, risus quis bibendum facilisis, lorem purus rutrum neque, nec porta tortor urna quis orci. Aenean aliquet, libero semper volutpat luctus, pede erat lacinia augue, quis rutrum sem ipsum sit amet pede. Vestibulum aliquet, nibh sed iaculis sagittis, odio dolor blandit augue, eget mollis urna tellus id tellus. Aenean aliquet aliquam nunc. Nulla ultricies justo eget orci. Phasellus tristique fermentum leo. Sed massa metus, sagittis ut, semper ut, pharetra vel, erat. Aliquam quam turpis, egestas vel, elementum in, egestas sit amet, lorem. Duis convallis, wisi sit amet mollis molestie, libero mauris porta dui, vitae aliquam arcu turpis ac sem. Aliquam aliquet dapibus metus.

Vivamus commodo eros eleifend dui. Vestibulum in leo eu erat tristique mattis. Cras

at elit. Cras pellentesque. Nullam id lacus sit amet libero aliquet hendrerit. Proin placerat, mi non elementum laoreet, eros elit tincidunt magna, a rhoncus sem arcu id odio. Nulla eget leo a leo egestas facilisis. Curabitur quis velit. Phasellus aliquam, tortor nec ornare rhoncus, purus urna posuere velit, et commodo risus tellus quis tellus. Vivamus leo turpis, tempus sit amet, tristique vitae, laoreet quis, odio. Proin scelerisque bibendum ipsum. Etiam nisl. Praesent vel dolor. Pellentesque vel magna. Curabitur urna. Vivamus congue urna in velit. Etiam ullamcorper elementum dui. Praesent non urna. Sed placerat quam non mi. Pellentesque diam magna, ultricies eget, ultrices placerat, adipiscing rutrum, sem.

Morbi sem. Nulla facilisi. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Nulla facilisi. Morbi sagittis ultrices libero. Praesent eu ligula sed sapien auctor sagittis. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Donec vel nunc. Nunc fermentum, lacus id aliquam porta, dui tortor euismod eros, vel molestie ipsum purus eu lacus. Vivamus pede arcu, euismod ac, tempus id, pretium et, lacus. Curabitur sodales dapibus urna. Nunc eu sapien. Donec eget nunc a pede dictum pretium. Proin mauris. Vivamus luctus libero vel nibh.

Fusce tristique risus id wisi. Integer molestie massa id sem. Vestibulum vel dolor. Pellentesque vel urna vel risus ultricies elementum. Quisque sapien urna, blandit nec, iaculis ac, viverra in, odio. In hac habitasse platea dictumst. Morbi neque lacus, convallis vitae, commodo ac, fermentum eu, velit. Sed in orci. In fringilla turpis non arcu. Donec in ante. Phasellus tempor feugiat velit. Aenean varius massa non turpis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae;

Aliquam tortor. Morbi ipsum massa, imperdiet non, consectetur vel, feugiat vel, lorem. Quisque eget lorem nec elit malesuada vestibulum. Quisque sollicitudin ipsum vel sem. Nulla enim. Proin nonummy felis vitae felis. Nullam pellentesque. Duis rutrum feugiat felis. Mauris vel pede sed libero tincidunt mollis. Phasellus sed urna rhoncus diam euismod bibendum. Phasellus sed nisl. Integer condimentum justo id orci iaculis varius. Quisque et lacus. Phasellus elementum, justo at dignissim auctor, wisi odio lobortis arcu, sed sollicitudin felis felis eu neque. Praesent at lacus.

Vivamus sit amet pede. Duis interdum, nunc eget rutrum dignissim, nisl diam luctus leo, et tincidunt velit nisl id tellus. In lorem tellus, aliquet vitae, porta in, aliquet sed, lectus. Phasellus sodales. Ut varius scelerisque erat. In vel nibh eu eros imperdiet rutrum. Donec ac odio nec neque vulputate suscipit. Nam nec magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nullam porta, odio et sagittis iaculis, wisi neque fringilla sapien, vel commodo lorem lorem id elit. Ut sem lectus, scelerisque eget, placerat et, tincidunt scelerisque, ligula. Pellentesque non orci.

Etiam vel ipsum. Morbi facilisis vestibulum nisl. Praesent cursus laoreet felis. Integer adipiscing pretium orci. Nulla facilisi. Quisque posuere bibendum purus. Nulla quam mauris, cursus eget, convallis ac, molestie non, enim. Aliquam congue. Quisque sagittis nonummy sapien. Proin molestie sem vitae urna. Maecenas lorem. Vivamus viverra consequat enim.

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Appendix

Sem můžete přidat přílohu. Pokud chcete “Přílohy”, tak upravte definici záhlaví v souboru sci.muni.thesis.sty, viz příkaz \HlavickaPriloha.

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